1. Novatel: KVH-1750 IMU
    - Use the "NovAtel/1.RAWIMU.mat" file:
        - **f** → scaled forces (accelerometers) with components **x,y,z** (m/s^2)
        - **w** → angular rotations (gyros) with components **x,y,z** in (rad/s)
        - **IMU_second** for **synchronization purposes**
        - **Note 1** the KVH sensors data from the y axis are inverted, so you have to invert the readings simply by multiplying by (-1)

```
fy = f.y * -1;
wy = w.y * -1;
```

        - **Note 2** The IMU data frequency is 200 Hz so you will need to down sample the data for the 3D-RISS to 1 Hz (the odometer data frequency)
2. TPI: VTI IMU
    - Use the "TPI/ TPI_data_interpolated2.mat" file:
        - **f** → scaled forces (accelerometers) with components **x,y,z** (m/s^2)
        - **w** → angular rotations (gyros) with components (rad/s)
        - **IMU_second** for **synchronization purposes**
        - **Note** The IMU data frequency is 20 Hz so you will need to down sample the data for the 3D-RISS to 1 Hz (the odometer data frequency)
3. Odometer data
    - Use the "OBDII_data/CarChip_Speed.mat" for the speed data in (m/s) with 1 Hz frequency.
    - Use the "OBDII_data/odo_second.mat" for synchronization purposes.

4. Reference data
    - Use the "NovAtel/3.INSPVA_Reference.mat" file:
        - Position components are **INS_Long, INS_Lat, INS_Alt**
        - Velocity components are **INS_ve, INS_vn, INS_vu**
        - Orientation components are **INS_Pitch, INS_Roll, INS_Azi**
        - **INS_second** for **synchronization purposes**
        - **Note 1** **INS** here means the **Integrated Navigation Solution**
        - **Note 2** The reference data frequency is 5 Hz so you will need to down sample that to compare with your 3D-RISS solution (1 Hz)
        - **Note 3** The reference solution angular components are in degrees so you could convert them to radians if you want to compare with your solution or vise versa. **INS_Long, INS_Lat, INS_Pitch, INS_Roll, INS_Azi**
        - **Note 4** The reference solution is to be used to obtain the initial navigation states (Position, Velocity, Attitude) for the RISS solution.